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Thermodynamic modelling of the cerium–oxygen system☆

Matvei Zinkevich*, Dejan Djurovic, Fritz Aldinger

Max-Planck Institut für Metallforschung, Institut für Nichtmetallische Anorganische Materialien, Universität Stuttgart, Heisenbergstr. 3, D-70569 Stuttgart, Germany

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Abstract

The published data for the thermochemical properties and phase relationships of the cerium–oxygen system have been critically examined. No major inconsistencies in the experimental database have been found. Thermodynamic model parameters of all phases, i.e., liquid, fcc-Ce, bcc-Ce, A-, H-, and X-Ce_2O_{3-x}, $Ce_3O_{5\pm x}$ (Mn_2O_3-type), CeO_{2-x}, as well as $Ce_{11}O_{20}$, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$, $Ce_{19}O_{34}$, Ce_9O_{16}, and Ce_7O_{12} have been derived by the least-squares minimization procedure using Thermo-Calc® software. The models are simple to be compatible with the standard Gibbs energy minimization procedure, but provide enough flexibility to account for the crystal structure, defect chemistry, and thermodynamic properties of phases in the Ce–O system. The backward compatibility of the refined parameters with experimental data has been demonstrated by calculation of phase and property diagrams. The Ce–O phase diagram in the whole range of compositions and at elevated temperature, a complete list of invariant reactions as well as the enthalpy and entropy of formation of Ce_nO_{2n-2m} phases are presented for the first time.
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Keywords: Ceria; Phase diagram; Thermodynamics; Defect chemistry; Compound energy formalism

1. Introduction

Owing to the great technological interest, cerium–oxygen system has been a subject of numerous studies over the past several decades. Cerium oxides are of interest to the steel industry since cerium is used as a deoxidizing and desulphurising agent and also improve the oxidation resistance of stainless steel. Pure ceria is an effective catalyst, while it becomes solid electrolyte upon doping with cations of lower valency. The phase diagram of Ce–O system, however, is still a subject of controversy. On the one hand, this is due to the complexity of phase relations at low temperatures, where the defect fluorite structure orders to form a series of discrete compounds Ce_nO_{2n-2m}. On the other hand, cerium oxides are refractory compounds, so that the liquid phase is difficult to study experimentally.

Considerable efforts have been dedicated to the model representation of the nonstoichiometry of cerium dioxide using the CALPHAD method [1–3] and statistical–thermodynamic approach [4–7]. However, none of the proposed models accounts simultaneously for thermochemical properties, the defect chemistry and the Ce–O phase diagram in the whole composition range. The development of a model, which does so is the objective of the present work.

2. Survey of literature information

The binary system Ce–O contains the gas phase, the liquid phase, the interstitial solid solutions of oxygen in fcc- and bcc-Ce and various oxide phases (Table 1). The dioxide, CeO_2 and sesquioxide, Ce_2O_3 are well-known compounds, whereas a confusion existed in the literature regarding intermediate phases and their thermodynamic stability [12]. Recent studies [9,13–15] have shown that a variety of discrete compounds, which have been reported to exist in the Ce–O system are commensurate superstructures of the fluorite-type lattice and their compositions can be rationalized as Ce_nO_{2n-2m}, where n and m are integers ($n>m$). Apart from the phases shown in

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* Corresponding author. Tel.: +49 711 68 93 105; fax: +49 711 68 93 131.
E-mail address: zinkevich@mf.mpg.de (M. Zinkevich).
URL: http://aldix.mpi-stuttgart.mpg.de/zinkevit/home_mz.html (M. Zinkevich).

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Table 1
Crystallographic data for cerium oxides

Phase	O/Ce ratio	n	m	Pearson symbol	Space group	Prototype	Lattice parameters[a]	Ref.
Ce_2O_{3-x} (A)	$(3-x)/2$	4	1	$hP5$	$P\bar{3}2/m1$	A-La_2O_3	a=0.3891(1) nm c=0.6059(1) nm	[8]
$Ce_3O_{5\pm x}$ (C)	$(5\pm x)/3$	6	1	$cI96$	$la\bar{3}$	Mn_2O_3	a=1.111(2) nm	[9]
Ce_7O_{12}	1.714	7	1	$hR19$	$R\bar{3}$	UY_6O_{12}	a=0.6785(1) nm α=99.42(1)°	[9]
Ce_9O_{16}	1.778	9	1	–	–	–	–	[10]
$Ce_{19}O_{34}$	1.790	19	2	–	triclinic	–	a=0.6627 nm b=1.1478 nm c=1.0123 nm α=100.9° β=90.0° γ=95.5°	[11]
$Ce_{40}O_{72}$	1.800	40	4	–	–	–	–	[10]
$Ce_{62}O_{112}$	1.806	62	6	–	–	–	–	[11]
$Ce_{11}O_{20}$	1.818	11	1	$aP31$	$P\bar{1}$	$Tb_{11}O_{20}$	a=0.6757(3) nm b=1.0260(5) nm c=0.6732(3) nm α=90.04(4)° β=99.80(4)° γ=96.22(4)°	[9]
CeO_{2-x} (F)	$2-x$	$\propto$	1	$cF12$	$Fm\bar{3}m$	CaF_2	a=5.4112(10) nm	[9]

The numbers n and m refer to the stoichiometric compounds Ce_nO_{2n-2m}.
[a] For the stoichiometric composition (x=0).

Table 1, compounds with n=13, 29 and 39 may also exist [11], but neither their chemical composition nor the stability range is known. In the present work, the following phase designations are used: F for CeO_{2-x}, C for $Ce_3O_{5\pm x}$ and A for Ce_2O_{3-x}. The latter exhibits two reversible polymorphic transformations, similar to La_2O_3 [16]. Hexagonal A-Ce_2O_3 transforms into partially ordered hexagonal H-Ce_2O_3 at 2338 K, then into cubic X-Ce_2O_3 at 2413 K [17,18]. These high-temperature modifications are designated as H and X. Discrete Ce_nO_{2n-2m} compounds are identified by the stoichiometric numbers.

The experimental database for Ce–O system (Table 2) includes results from measurements of equilibrium oxygen partial pressure (direct or by electromotive force method), partial enthalpy of oxygen dissolution, enthalpy of formation, enthalpy of invariant reactions, and heat capacity or enthalpy increment over a wide range of compositions and temperatures as well as determinations of phase boundaries in the region of compositions between Ce_2O_{3-x} and CeO_2 and close to metallic Ce. Almost all these data have already been reviewed [1,12,42,64,65] and only a summary is given here. Both the subregular solution model [1,3] and the sublattice model [2] were successfully applied to describe thermodynamic properties of the nonstoichiometric ceria (CeO_{2-x}). Lindemer [1] analysed the differences between observed and calculated oxygen potentials for a given value of x and temperature. With the exception of three datasets [23,25,28], for which the residuals reach the value of 100 kJ·mol^{-1}, the experimental database was found to show good consistency.

The reported melting temperatures of Ce_2O_3 and CeO_2 show considerable scatter (Table 3). Obviously, the scattering is due to the fact that both compounds become oxygen deficient at high temperatures and neither the type of melting (congruent or incongruent) nor the actual composition could be determined. The data on the melting temperature of Ce_2O_3 were reviewed by Coutures and Rand [75]. Based on the careful analysis of the measurements conditions, T_m=2503±50 K was recommended as probable value and it is accepted in the present work. Watson [29] evaluated the literature, where the melting temperature of CeO_2 is given. He recommended a value of 2753 K measured by Foex [74] in air using a solar furnace. However, since ceria is very prone to reduction, it can be expected that stoichiometric CeO_2 will melt congruently only under elevated pressure of oxygen. Indeed, Mordovin et al. [67] detected the liquid phase already at 2670 K, when heating ceria in an argon atmosphere. The available surveys [69–73] give a higher melting point of ceria, in the range 2998 to 3073 K, but the only value, for which the confidence interval is given is T_m=3000±20 K [70], and it is accepted in this work.

3. Thermodynamic models

With the CALPHAD-method [76,77], all available thermodynamic and phase equilibrium data for the system are evaluated simultaneously to obtain one set of model equations for the Gibbs energies of all phases as functions of temperature and composition. Some of these expressions contain adjustable coefficients (model parameters). The optimal values of the unknown parameters providing the best match between the calculated quantities and their experimental counterparts are usually obtained by the weighted non-linear least-squares minimization procedure (thermodynamic optimization). From these equations, the thermodynamic properties and the phase diagrams can be back-calculated. The selection of the model for a phase must be based on the physical and chemical properties

Table 2
The experimental database for Ce–O system

Data description	Temperature range (K)	Composition range (O/Ce)	Phases involved	Ref.
Equilibrium oxygen pressure	813–1323	1.670–1.996	F, A	[19]
Equilibrium oxygen pressure	968–1282	1.581–1.960	F, A	[20]
Equilibrium oxygen pressure	909–1442	1.500–1.997	F, C, A, Ce_7O_{12}	[21]
Equilibrium oxygen pressure	1073–1273	1.930–1.984	F	[22]
Equilibrium oxygen pressure	1173–1573	1.727–2.000	F	[23]
Equilibrium oxygen pressure	1173–1323	1.710–2.000	F	[24]
Equilibrium oxygen pressure	1173–1823	1.723–1.992	F	[25]
Equilibrium oxygen pressure	900–1400	1.650–2.000	F, C, A, Ce_7O_{12}	[26]
Equilibrium oxygen pressure	1023–1773	1.721–2.000	F	[27]
Equilibrium oxygen pressure	1173–1673	1.749–1.997	F	[28]
Equilibrium oxygen pressure	1750–2243	1.600–1.958	F, liquid	[29]
Equilibrium oxygen pressure	1244–1353	1.731–1.999	F, C, A, Ce_7O_{12}	[30]
Equilibrium oxygen pressure	1273–1573	1.720–2.000	F	[31]
Equilibrium oxygen pressure	1073	1.997–2.000	F	[32]
Equilibrium oxygen pressure	1273–1603	1.647–1.994	F, C, A	[33]
Equilibrium oxygen pressure	1073–1218	1.968–1.998	F	[34]
Equilibrium oxygen pressure	1073	1.995–2.000	F	[35]
Equilibrium oxygen pressure	1073–1273	1.750–2.000	F	[36]
Partial enthalpy of oxygen dissolution	1353	1.529–1.994	F, C, A	[37]
Enthalpy of formation	298	1.500	A	[38–42]
Heat capacity	50–298	1.500	A	[43]
Heat capacity	5–300	1.500	A	[44]
Heat capacity	3–420	1.500	A	[42]
Heat capacity	578–1116	1.500	A	[45]
Heat capacity	400–1100	1.500	A	[46]
Enthalpy increment	400–1000	1.500	A	[47]
Enthalpy increment	470–883	1.500	A	[42]
Enthalpy of formation	298	2.000	F	[48,49]
Heat capacity	5–300	2.000	F	[50]
Heat capacity	608–1172	2.000	F	[51]
Heat capacity	418–758	2.000	F	[52]
Enthalpy increment	298–1800	2.000	F	[53]
Enthalpy increment	492–1141	2.000	F	[54]
Enthalpy increment	391–1624	2.000	F	[55]
Enthalpy of reaction	722	1.846	F, $Ce_{11}O_{20}$	[56]
Enthalpy of reaction	736	1.818	F, $Ce_{11}O_{20}$, $Ce_{62}O_{112}$	[56]
Enthalpy of reaction	766	1.808	F, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$	[56]
Heat capacity	348–1200	1.823–2.000	F, $Ce_{11}O_{20}$, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$	[57]
Phase diagram	293–1273	1.790–2.000	F	[58]
Phase diagram	673–901	1.800–2.000	F	[59]
Phase diagram	573–1073	1.750–1.960	F, $Ce_{11}O_{20}$, Ce_9O_{16}, Ce_7O_{12}	[60]
Phase diagram	600–1200	1.714–2.000	F, $Ce_{11}O_{20}$, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$, $Ce_{19}O_{34}$, Ce_9O_{16}, Ce_7O_{12}	[56,57,61]
Phase diagram	597–1064	1.698–1.800	F, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$, $Ce_{19}O_{34}$, Ce_9O_{16}, Ce_7O_{12}	[15]
Phase diagram	2123–2243	1.600–1.660	F, liquid	[29]
Phase diagram	1606–2302	1.330–1.540	A, liquid	[62]
Phase diagram	923–1386	0.002–0.008	fcc-Ce, bcc-Ce, liquid, A	[63]

of that phase, most importantly crystallography, type of bonding, ordering, and defect structure. In this way, all the data are rendered self-consistent and consistent with thermodynamic principles. In this work, the thermodynamic optimization and calculations were carried out using the software package "Thermo-Calc" [78].

Since the enthalpies cannot be defined absolutely, the Gibbs energy is referred to the constant enthalpy values of the so-called Stable Element References, H^{SER}, at 298.15 K and 1 bar [79]. The Gibbs energy of the pure elements is taken from the SGTE unary database [79]. The Gibbs energy of stoichiometric solid phases ($Ce_{11}O_{20}$, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$, $Ce_{19}O_{34}$, Ce_9O_{16}, Ce_7O_{12}) and end-members of solutions (see below) are represented as a power series in terms of temperature in the form [79]

$$^{\circ}G(T) = a + bT + cT\ln(T) + \sum_n g_n T^n, \quad (1)$$

where a to f and g_n are coefficients and n represents a set of integers, typically taking the values of 2, 3 and −1.

The liquid phase was described by the two-sublattice partially ionic liquid model [80,81] with trivalent and tetravalent cerium on the cation sublattice and divalent oxygen and vacancies on the anion sublattice, i.e., $(Ce^{3+}, Ce^{4+})_p$ $(Va^{q-}, O^{2-})_q$, where the

Table 3
Melting points reported for CeO_2 and Ce_2O_3

Phase	Melting point (K)	Environment	Ref.	Year
Ce_2O_3	2433±30	–	[66]	1966
Ce_2O_3	2415±30	H_2 in tungsten tube	[67]	1967
Ce_2O_3	2483±10	H_2	[68]	1968
Ce_2O_3	2513±20	Ar in molybdenum crucible	[17]	1971
Ce_2O_3	2513±20	Ar in molybdenum crucible	[18]	1985
CeO_2	3073	Under pressure (16 atm)	[69]	1949
CeO_2	3000±20	Under O_2-pressure	[70]	1953
CeO_2	3023	–	[71]	1960
CeO_2	2998	–	[72]	1961
CeO_2	3023	–	[73]	1964
CeO_2	2753	Air in solar furnace	[74]	1966
CeO_2	2670	Ar in tungsten tube	[67]	1967

number of sites on the respective sublattice, p and q, vary with composition in order to maintain electroneutrality. With this model, it is possible to handle the whole range of compositions from pure cerium to CeO_2 and to describe a gradual change from metallic to ionic liquid. The molar Gibbs energy is given by

$$\begin{aligned} G^{\mathrm{liq}} = {} & y_{\mathrm{Ce^{3+}}} y_{\mathrm{O^{2-}}} {}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce_2O_3}} + y_{\mathrm{Ce^{4+}}} y_{\mathrm{O^{2-}}} 2{}^{\circ}G^{\mathrm{liq}}_{\mathrm{CeO_2}} \\ & + q\left(y_{\mathrm{Ce^{3+}}} y_{\mathrm{Va^{q-}}} {}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{3+}:Va^{q-}}} + y_{\mathrm{Ce^{4+}}} y_{\mathrm{Va^{q-}}} {}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{4+}:Va^{q-}}}\right) \\ & + pRT\left(y_{\mathrm{Ce^{3+}}} \ln y_{\mathrm{Ce^{3+}}} + y_{\mathrm{Ce^{4+}}} \ln y_{\mathrm{Ce^{4+}}}\right) \\ & + qRT\left(y_{\mathrm{O^{2-}}} \ln y_{\mathrm{O^{2-}}} + y_{\mathrm{Va^{q-}}} \ln y_{\mathrm{Va^{q-}}}\right) + {}^{\mathrm{E}}G^{\mathrm{liq}}, \end{aligned} \tag{2}$$

where y_i represents the site fraction of the species i on the respective sublattice, while ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce_2O_3}}$ and ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{CeO_2}}$ are equal to the Gibbs energy of liquid Ce_2O_3 and CeO_2 respectively. The parameter ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{4+}:Va^{q-}}}$ represents pure liquid Ce with Ce^{4+} ions, whereas real liquid Ce is represented by the parameter ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{3+}:Va^{q-}}}$ (i.e., with Ce^{3+} ions). The fraction of Ce^{4+} ions in the liquid should be very small for O/Ce<1.5, so that a large positive value should be given to ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{4+}:Va^{q-}}}$ relative to ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{3+}:Va^{q-}}}$. In this work it is achieved using the reciprocal relation

$${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{4+}:Va^{q-}}} + 2{}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce_2O_3}} = 2{}^{\circ}G^{\mathrm{liq}}_{\mathrm{Ce^{3+}:Va^{q-}}} + 3{}^{\circ}G^{\mathrm{liq}}_{\mathrm{CeO_2}}. \tag{3}$$

A similar formalism was already used to describe the liquid in the Cu–O system [82]. The remaining term in Eq. (2), ${}^{\mathrm{E}}G^{\mathrm{liq}}$, is the excess Gibbs energy, which can be expressed in terms of interaction parameters (L) using Redlich–Kister type polynomial series [83]:

$$\begin{aligned} {}^{\mathrm{E}}G^{\mathrm{liq}} = {} & y_{\mathrm{Ce^{3+}}} y_{\mathrm{O^{2-}}} y_{\mathrm{Va^{q-}}} \sum_{\nu=0}^{n} {}^{\nu}L^{\mathrm{liq}}_{\mathrm{Ce^{3+}:O^{2-},Va^{q-}}} \left(y_{\mathrm{O^{2-}}} - y_{\mathrm{Va^{q-}}}\right)^{\nu} \\ & + y_{\mathrm{Ce^{4+}}} y_{\mathrm{O^{2-}}} y_{\mathrm{Va^{q-}}} \sum_{\nu=0}^{n} {}^{\nu}L^{\mathrm{liq}}_{\mathrm{Ce^{4+}:O^{2-},Va^{q-}}} \left(y_{\mathrm{O^{2-}}} - y_{\mathrm{Va^{q-}}}\right)^{\nu} \\ & + y_{\mathrm{Ce^{3+}}} y_{\mathrm{Ce^{4+}}} y_{\mathrm{O^{2-}}} \sum_{\nu=0}^{n} {}^{\nu}L^{\mathrm{liq}}_{\mathrm{Ce^{3+},Ce^{4+}:O^{2-}}} \left(y_{\mathrm{Ce^{3+}}} - y_{\mathrm{Ce^{4+}}}\right)^{\nu} \\ & + y_{\mathrm{Ce^{3+}}} y_{\mathrm{Ce^{4+}}} y_{\mathrm{Va^{q-}}} \sum_{\nu=0}^{n} {}^{\nu}L^{\mathrm{liq}}_{\mathrm{Ce^{3+},Ce^{4+}:Va^{q-}}} \left(y_{\mathrm{Ce^{3+}}} - y_{\mathrm{Ce^{4+}}}\right)^{\nu} \end{aligned} \tag{4}$$

Commas separate species on the same sublattice, and colons separate species on different sublattices. The thermodynamic properties of liquid in the metal-rich region (O/Ce<1.5) are described by the interaction parameter ${}^{\nu}L^{\mathrm{liq}}_{\mathrm{Ce^{3+}:O^{2-},Va^{q-}}}$, while the interactions in the range 1.5<O/Ce<2.0 are described by the

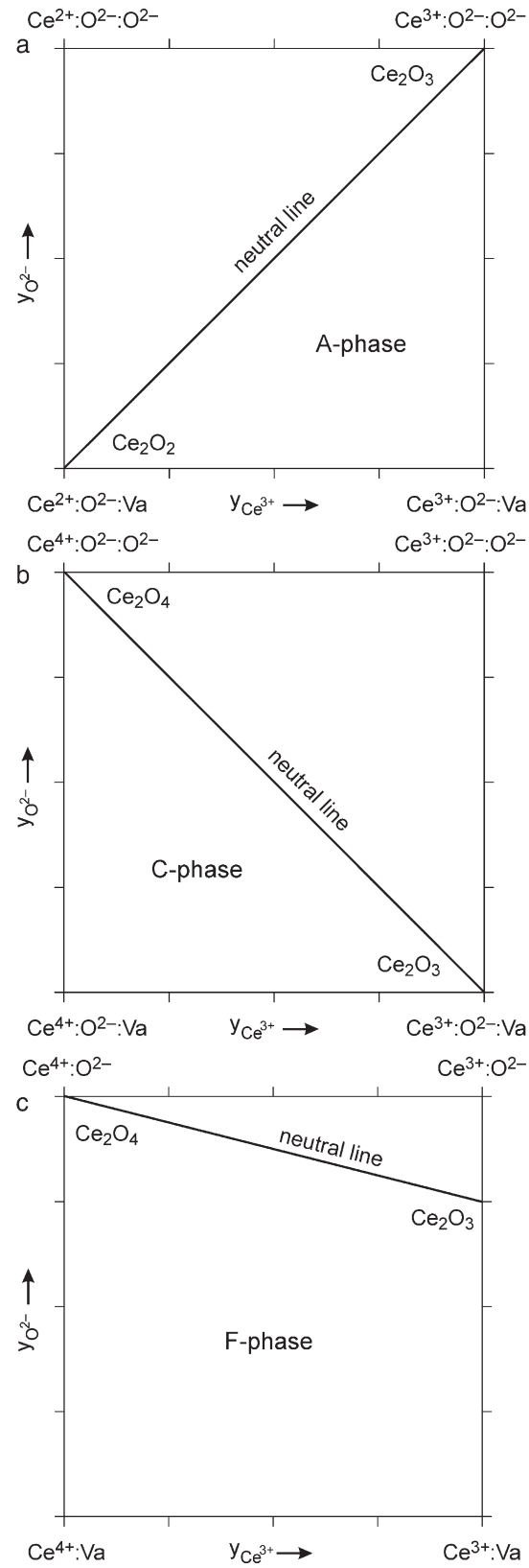

Fig. 1. Geometric representation of thermodynamic models for the nonstoichiometric cerium oxide phases: Ce_2O_{3-x} (a), $Ce_3O_{5\pm x}$ (b), and CeO_{2-x} (c).

Table 4
Summary of the thermodynamic parameters describing condensed phases in the Ce–O system referred to stable element reference H^{SER}(T=298.15K, P=1 bar)

Phase, parameter	Ref.
Liquid $(Ce^{3+}, Ce^{4+})_p$ $(Va^{q-}, O^{2-})_q$, Eqs. (2)–(4)	
$^{\circ}G^{liq}_{Ce_2O_3} = 126654-54.497\,T + {}^{\circ}G_{A-Ce_2O_3}$	This work
$^{\circ}G^{liq}_{CeO_2} = 87177-29.059\,T + {}^{\circ}G_{CeO_2}$	This work
$^{\circ}G^{liq}_{Ce^{3+}:Va^{q-}} = {}^{\circ}G_{Ce(l)}$	[79]
$^{\circ}G^{liq}_{Ce^{4+}:Va^{q-}} = 2^{\circ}G_{Ce(l)}+3^{\circ}G^{liq}_{CeO_2}-2^{\circ}G^{liq}_{Ce_2O_3}$	This work
$^{0}L^{liq}_{Ce^{3+},Ce^{4+}:O^{2-}} = -105398$	This work
$^{1}L^{liq}_{Ce^{3+},Ce^{4+}:O^{2-}} = -35158$	This work
$^{0}L^{liq}_{Ce^{3+}:O^{2-},Va^{q-}} = 114250$	This work
fcc (bcc)-Ce $(Ce)_1(O,Va)_1$, Eq. (5)	
$^{\circ}G^{fcc}_{Ce:Va} = {}^{\circ}G_{Ce(fcc)}$	[79]
$^{\circ}G^{fcc}_{Ce:O} = -525009 + 67.7626T + {}^{\circ}G_{Ce(fcc)} + 1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{bcc}_{Ce:Va} = {}^{\circ}G_{Ce(bcc)}$	[79]
$^{\circ}G^{bcc}_{Ce:O} = -521242 + 67.7626T + {}^{\circ}G_{Ce(bcc)} + 1/2^{\circ}G_{O_2(g)}$	This work
A-phase $(Ce^{3+},Ce^{2+})_2(O^{2-})_2(O^{2-},Va)_1$, Eqs. (6)–(8)	
$^{\circ}G^{A}_{Ce^{3+}:O^{2-}:O^{2-}} = {}^{\circ}G_{A-Ce_2O_3}$	This work
$^{\circ}G^{A}_{Ce^{3+}:O^{2-}:Va} = {}^{\circ}G_{A-Ce_2O_3}-1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{A}_{Ce^{2+}:O^{2-}:Va} = {}^{\circ}G_{Ce_2O_2}$	This work
$^{\circ}G^{A}_{Ce^{2+}:O^{2-}:O^{2-}} = {}^{\circ}G_{Ce_2O_2} + 1/2^{\circ}G_{O_2(g)}$	This work
$^{0}L^{A}_{Ce^{3+},Ce^{2+}:O^{2-}:O^{2-}} = {}^{0}L^{A}_{Ce^{3+},Ce^{2+}:O^{2-}:Va} = 102206$	This work
H-phase $(Ce^{3+},Ce^{2+})_2(O^{2-})_2(O^{2-},Va)_1$, Eqs. (6)–(8)	
$^{\circ}G^{H}_{Ce^{3+}:O^{2-}:O^{2-}} = {}^{\circ}G_{H-Ce_2O_3}$	[86]
$^{\circ}G^{H}_{Ce^{3+}:O^{2-}:Va} = {}^{\circ}G_{H-Ce_2O_3}-1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{H}_{Ce^{2+}:O^{2-}:Va} = {}^{\circ}G_{Ce_2O_2}$	This work
$^{\circ}G^{H}_{Ce^{2+}:O^{2-}:O^{2-}} = {}^{\circ}G_{Ce_2O_2} + 1/2^{\circ}G_{O_2(g)}$	This work
$^{0}L^{H}_{Ce^{3+},Ce^{2+}:O^{2-}:O^{2-}} = {}^{0}L^{H}_{Ce^{3+},Ce^{2+}:O^{2-}:Va} = 102206$	This work
X-phase $(Ce^{3+},Ce^{2+})_2(O^{2-})_2(O^{2-},Va)_1$, Eqs. (6)–(8)	
$^{\circ}G^{X}_{Ce^{3+}:O^{2-}:O^{2-}} = {}^{\circ}G_{X-Ce_2O_3}$	[86]
$^{\circ}G^{X}_{Ce^{3+}:O^{2-}:Va} = {}^{\circ}G_{X-Ce_2O_3}-1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{X}_{Ce^{2+}:O^{2-}:Va} = {}^{\circ}G_{Ce_2O_2}$	This work
$^{\circ}G^{X}_{Ce^{2+}:O^{2-}:O^{2-}} = {}^{\circ}G_{Ce_2O_2} + 1/2^{\circ}G_{O_2(g)}$	This work
$^{0}L^{X}_{Ce^{3+},Ce^{2+}:O^{2-}:O^{2-}} = {}^{0}L^{X}_{Ce^{3+},Ce^{2+}:O^{2-}:Va} = 102206$	This work
C-phase $(Ce^{3+},Ce^{4+})_2(O^{2-})_3(O^{2-},Va)_1$, Eq. (9)	
$^{\circ}G^{C}_{Ce^{3+}:O^{2-}:Va} = {}^{\circ}G_{C-Ce_2O_3}$	[86]
$^{\circ}G^{C}_{Ce^{3+}:O^{2-}:O^{2-}} = {}^{\circ}G_{C-Ce_2O_3} + 1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{C}_{Ce^{4+}:O^{2-}:Va} = 19140 + 2^{\circ}G_{CeO_2}-1/2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{C}_{Ce^{4+}:O^{2-}:O^{2-}} = 19140 + 2^{\circ}G_{CeO_2}$	This work
$^{0}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:O^{2-}} = {}^{0}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:Va} = -7178-17.7264\,T$	This work
$^{1}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:O^{2-}} = {}^{1}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:Va} = -25591+22.2801\,T$	This work
F-phase $(Ce^{4+},Ce^{3+})_2(O^{2-},Va)_4$, Eqs. (10)–(12)	
$^{\circ}G^{F}_{Ce^{4+}:O^{2-}} = 2^{\circ}G_{CeO_2}$	This work
$^{\circ}G^{F}_{Ce^{4+}:Va} = 2^{\circ}G_{CeO_2}-2^{\circ}G_{O_2(g)}$	This work
$^{\circ}G^{F}_{Ce^{3+}:O^{2-}} = {}^{\circ}G_{F-Ce_2O_3} + 1/2^{\circ}G_{O_2(g)} + 18.702165\,T$	This work
$^{\circ}G^{F}_{Ce^{3+}:Va} = {}^{\circ}G_{F-Ce_2O_3}-1.5^{\circ}G_{O_2(g)} + 18.702165\,T$	This work
$^{0}L^{F}_{Ce^{3+},Ce^{4+}:O^{2-}} = {}^{0}L^{F}_{Ce^{4+},Ce^{3+}:Va} = -117572 + 25.9745\,T$	This work
$^{1}L^{F}_{Ce^{3+},Ce^{4+}:O^{2-}} = {}^{1}L^{F}_{Ce^{4+},Ce^{3+}:Va} = -115033 + 31.1026\,T$	This work
$^{\circ}G_{Ce_7O_{12}} = -133719-11.0961\,T + 3^{\circ}G_{CeO_2} + 2^{\circ}G_{F-Ce_2O_3}$	This work
$^{\circ}G_{Ce_9O_{16}} = -125351-31.333\,T + 5^{\circ}G_{CeO_2} + 2^{\circ}G_{F-Ce_2O_3}$	This work

Table 4 (*continued*)

Phase, parameter	Ref.
$^{\circ}G_{Ce_{19}O_{34}} = -247150-71\,T + 11^{\circ}G_{CeO_2} + 4^{\circ}G_{F-Ce_2O_3}$	This work
$^{\circ}G_{Ce_{40}O_{72}} = -488650-155\,T + 24^{\circ}G_{CeO_2} + 8^{\circ}G_{F-Ce_2O_3}$	This work
$^{\circ}G_{Ce_{62}O_{112}} = -730900-237.3\,T + 38^{\circ}G_{CeO_2} + 12^{\circ}G_{F-Ce_2O_3}$	This work
$^{\circ}G_{Ce_{11}O_{20}} = -121419-40.45\,T + 7^{\circ}G_{CeO_2} + 2^{\circ}G_{F-Ce_2O_3}$	This work
Functions	**Ref.**
$^{\circ}G_{A\text{-}Ce_2O_3}=-1832858+667.3307\,T-119.855\,T\ln T-0.012155\,T^2+546000\,T^{-1}$	This work
$^{\circ}G_{H\text{-}Ce_2O_3}=32699-13.986\,T+{}^{\circ}G_{A\text{-}Ce_2O_3}$	[86]
$^{\circ}G_{X\text{-}Ce_2O_3}=43724-18.555\,T+{}^{\circ}G_{A\text{-}Ce_2O_3}$	[86]
$^{\circ}G_{CeO_2}=-1116518+434.8839\,T-72.866\,T\ln T-0.003992\,T^2+602000\,T^{-1}$	This work
$^{\circ}G_{Ce_2O_2}=2/3^{\circ}G_{A\text{-}Ce_2O_3}+2/3^{\circ}G_{Ce(fcc)}$	This work
$^{\circ}G_{C\text{-}Ce_2O_3}=4390+7.232\,T+{}^{\circ}G_{A\text{-}Ce_2O_3}$	[86]
$^{\circ}G_{F\text{-}Ce_2O_3}=-1804481+1067.043\,T-175.3336\,T\ln T-6.498627\mathrm{E}-4\,T^2++546000\,T^{-1}$	This work
$^{\circ}G_{Ce(l)}=4117.865-11.423898\,T-7.5383948\,T\ln T-0.02936407\,T^2+4.827734\mathrm{E}-6\,T^3-198834\,T^{-1}$ (298 K<T<1000 K) $-6730.605+183.023193\,T-37.6978\,T\ln T$ (1000 K<T<4000 K)	[79]
$^{\circ}G_{Ce(fcc)}=-7160.519+84.23022\,T-22.3664\,T\ln T-0.0067103\,T^2-3.20773\mathrm{E}-7\,T^3-18117\,T^{-1}$ (298 K<T<1000 K) $-79678.506+659.4604\,T-101.32248\,T\ln T+0.026046487\,T^2-1.930297\mathrm{E}-6\,T^3+11531707\,T^{-1}$ (1000 K<T<2000 K) $-14198.639+190.370192\,T-37.6978\,T\ln T$ (2000 K<T<4000 K)	[79]
$^{\circ}G_{Ce(bcc)}=-1354.69-5.21501\,T-7.7305867\,T\ln T-0.029098402\,T^2+4.784299\mathrm{E}-6\,T^3-196303\,T^{-1}$ (298 K<T<1000 K) $-12101.106+187.449688\,T-37.6142\,T\ln T$ (1000 K<T<1072 K) $-11950.375+186.333811\,T-37.4627992\,T\ln T-5.7145\mathrm{E}-5\,T^2+2.348\mathrm{E}-9\,T^3-25897\,T^{-1}$ (1072 K<T<4000 K)	[79]

Values are given in SI units (Joule, mole, and Kelvin).

parameter $^{\nu}L^{liq}_{Ce^{3+},Ce^{4+}:O^{2-}}$. Both the interaction parameters $^{\nu}L^{liq}_{Ce^{4+}:O^{2-},Va^{q-}}$ and $^{\nu}L^{liq}_{Ce^{3+},Ce^{4+}:Va^{q-}}$ have only a minor influence and are set equal to zero for convenience.

The solid solution phases (fcc-Ce, bcc-Ce, A, C, and F) are described by a multi-sublattice model using the compound energy formalism [84]. Oxygen occupies octahedrally coordinated interstitial sites in solid Ce and both the fcc and bcc phase are then represented by a two-sublattice model $(Ce)_1(O,Va)_1$. Although there are three equivalent octahedral voids per metal atom in the bcc lattice, the occupation of an interstitial site is prevented by the prior occupation of a neighboring interstitial site, since the effective repulsion exists between near-neighbor O-atoms. This results in the splitting of the interstitial sites into three sublattices, while the maximum separation between oxygen atoms is achieved, if only one of them is occupied. The molar Gibbs energy is given by

$$G^{\varphi} = y_{Va}{}^{\circ}G^{\varphi}_{Ce:Va} + y_O{}^{\circ}G^{\varphi}_{Ce:O} + RT(y_O\ln y_O + y_{Va}\ln y_{Va}) + y_O y_{Va}{}^{0}L^{\varphi}_{Ce:O,Va}. \quad (5)$$

$^{\circ}G^{\varphi}_{Ce:Va}$ is the expression for the Gibbs energy of the pure fcc- or bcc-Ce, whereas $^{\circ}G^{\varphi}_{Ce:O}$ corresponds to the hypothetical

compound CeO and controls the oxygen solubility in solid cerium. The latter is low enough in order to assume zero for the interaction parameter ${}^{0}L^{\varphi}_{\mathrm{Ce:O,Va}}$.

Cerium sesquioxide (Ce_2O_3) is substoichiometric in equilibrium with cerium metal [62], while density and X-ray diffraction studies [85] indicate that substoichiometry is due to the formation of oxygen vacancies. The crystal structure of A-Ce_2O_3 can be described as hexagonal close packing of the cations, the oxygen ions, O1 and O2 respectively, occupying half of tetrahedral and half of octahedral holes [8]. Since the O2–Ce distance is longer than the O1–Ce one, it is reasonable to assume that the vacancies are formed on the octahedrally coordinated sites. In the model, the lack of negative charge in the anion sublattice due to oxygen vacancies is effectively compensated by the introduction of Ce^{2+} into the cation sublattice, i.e., $(Ce^{3+}, Ce^{2+})_2(O^{2-})_2(O^{2-},Va)_1$. Note that vacancies are neutral species here. The molar Gibbs energy of Ce_2O_{3-x} (A) is then represented as

$$\begin{aligned} G^{\mathrm{A}} = {} & y_{\mathrm{Ce^{3+}}} y_{\mathrm{O^{2-}}} {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:O^{2-}}} + y_{\mathrm{Ce^{3+}}} y_{\mathrm{Va}} {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:Va}} \\ & + y_{\mathrm{Ce^{2+}}} y_{\mathrm{O^{2-}}} {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:O^{2-}}} + y_{\mathrm{Ce^{2+}}} y_{\mathrm{Va}} {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:Va}} \\ & + 2RT\left(y_{\mathrm{Ce^{3+}}} \ln y_{\mathrm{Ce^{3+}}} + y_{\mathrm{Ce^{2+}}} \ln y_{\mathrm{Ce^{2+}}}\right) \\ & + RT\left(y_{\mathrm{O^{2-}}} \ln y_{\mathrm{O^{2-}}} + y_{\mathrm{Va}} \ln y_{\mathrm{Va}}\right) + {}^{\mathrm{E}}G^{\mathrm{A}} \end{aligned} \tag{6}$$

The model can be visualized by the composition square in Fig. 1a. Each corner represents a $^{\circ}G$-parameter. The end-member ${}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:O^{2-}}}$ corresponds to the stable form of A-Ce_2O_3. The condition of electroneutrality defines a line (Ce_2O_3–Ce_2O_2), on which all possible compositions fall. Ce_2O_2 is considered as a hypothetical compound, which keeps the crystal structure of A-Ce_2O_3, although all the O2-sites are vacant. Its molar Gibbs energy is expressed as

$${}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:Va}} = {}^{\circ}G_{\mathrm{Ce_2O_2}} = \frac{2}{3}{}^{\circ}G_{\mathrm{A-Ce_2O_3}} + \frac{2}{3}{}^{\circ}G_{\mathrm{Ce(fcc)}}. \tag{7}$$

The remaining end-members are charged and it would be impossible to give physically meaningful values to their Gibbs energies. The parameter ${}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:Va}}$ was chosen as a reference, i.e., ${}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:Va}} = {}^{\circ}G_{\mathrm{A-Ce_2O_3}} - 1/2\,{}^{\circ}G_{\mathrm{O_2(g)}}$, while the parameter ${}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:O^{2-}}}$ can be determined by the reciprocal relation, which does not involve the net change of charge:

$$\begin{aligned} {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:O^{2-}}} + {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:Va}} = {} & {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:Va}} \\ & + {}^{\circ}G^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:O^{2-}}}. \end{aligned} \tag{8}$$

The excess term, ${}^{\mathrm{E}}G^{\mathrm{A}}$ represents deviations from ideality and can be expressed in terms of interaction parameters (cf. Eq. (4)). The parameters ${}^{\nu}L^{\mathrm{A}}_{\mathrm{Ce^{3+},Ce^{2+}:O^{2-}:O^{2-}}}$ and ${}^{\nu}L^{\mathrm{A}}_{\mathrm{Ce^{3+},Ce^{2+}:O^{2-}:Va}}$ were used to model the extent of substoichiometry in Ce_2O_{3-x}, whereas ${}^{\nu}L^{\mathrm{A}}_{\mathrm{Ce^{3+}:O^{2-}:O^{2-},Va}}$ and ${}^{\nu}L^{\mathrm{A}}_{\mathrm{Ce^{2+}:O^{2-}:O^{2-},Va}}$ were not used and set equal to zero. Although the substoichiometry of the high-temperature modifications of Ce_2O_3 was not studied, there is no reason to expect significant differences compared to A-Ce_2O_3. Thus, the same model with the same values of ${}^{\circ}G_{\mathrm{Ce_2O_2}}$ and interaction parameters was used for A-, H-, and X-Ce_2O_{3-x}.

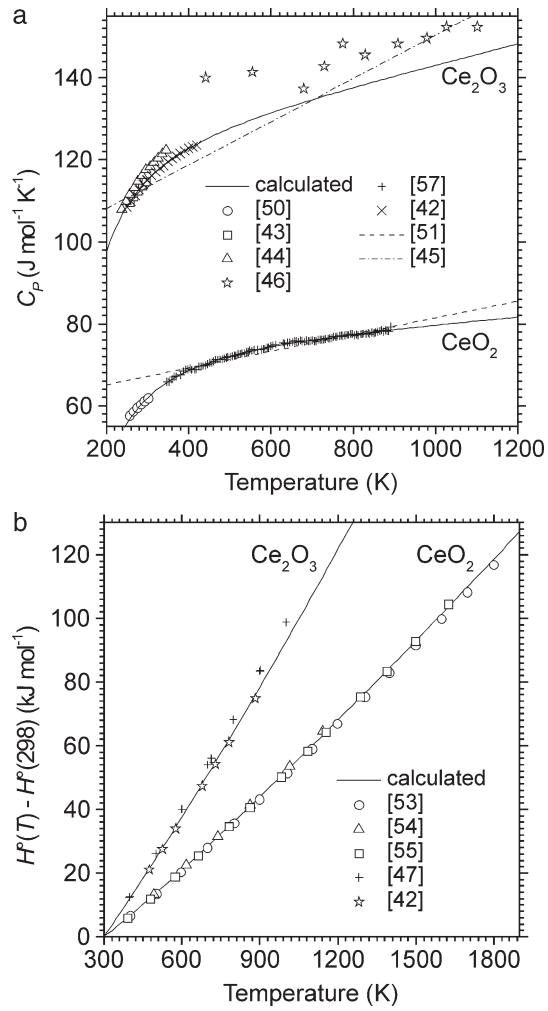

Fig. 2. Heat capacity (a) and enthalpy increment (b) of the stoichiometric Ce_2O_3 and CeO_2 as functions of temperature.

The crystal structure of $Ce_3O_{5\pm x}$ (C) contains four atomic positions in the unit cell. Cerium cations fully occupy 8*b*- and 24*d*-sites, while oxygen anions fully occupy 48*e*- and partly 16*c*-sites [9]. The variable oxygen content is effectively compensated by changing the oxidation state of cerium from 3+ to 4+. The sublattice formulation is $(Ce^{3+},Ce^{4+})_2(O^{2-})_3(O^{2-}, Va)_1$ and the molar Gibbs energy is given by

$$\begin{aligned} G^{\mathrm{C}} = {} & y_{\mathrm{Ce^{3+}}} y_{\mathrm{O^{2-}}} {}^{\circ}G^{\mathrm{C}}_{\mathrm{Ce^{3+}:O^{2-}:O^{2-}}} + y_{\mathrm{Ce^{3+}}} y_{\mathrm{Va}} {}^{\circ}G^{\mathrm{C}}_{\mathrm{Ce^{3+}:O^{2-}:Va}} \\ & + y_{\mathrm{Ce^{4+}}} y_{\mathrm{O^{2-}}} {}^{\circ}G^{\mathrm{C}}_{\mathrm{Ce^{4+}:O^{2-}:O^{2-}}} + y_{\mathrm{Ce^{4+}}} y_{\mathrm{Va}} {}^{\circ}G^{\mathrm{C}}_{\mathrm{Ce^{4+}:O^{2-}:Va}} \\ & + 2RT\left(y_{\mathrm{Ce^{3+}}} \ln y_{\mathrm{Ce^{3+}}} + y_{\mathrm{Ce^{4+}}} \ln y_{\mathrm{Ce^{4+}}}\right) \\ & + RT\left(y_{\mathrm{O^{2-}}} \ln y_{\mathrm{O^{2-}}} + y_{\mathrm{Va}} \ln y_{\mathrm{Va}}\right) + {}^{\mathrm{E}}G^{\mathrm{C}} \end{aligned} \tag{9}$$

The model can be visualized by the composition square in Fig. 1b. The end-member ${}^{\circ}G^{\mathrm{C}}_{\mathrm{Ce^{3+}:O^{2-}:Va}}$ corresponds to the cubic Ce_2O_3, which is metastable with respect to the A-form, while for heavier rare-earth sesquioxides and Y_2O_3 the cubic modifications are stable ones. Obviously, the ionic radius of Ce^{3+} is too large and explains the stabilisation of the C-phase by smaller Ce^{4+} cations. The other neutral end-member is Ce_2O_4, which contains tetravalent cerium only. Clearly, it must be unstable with respect to the fluorite-like CeO_2.

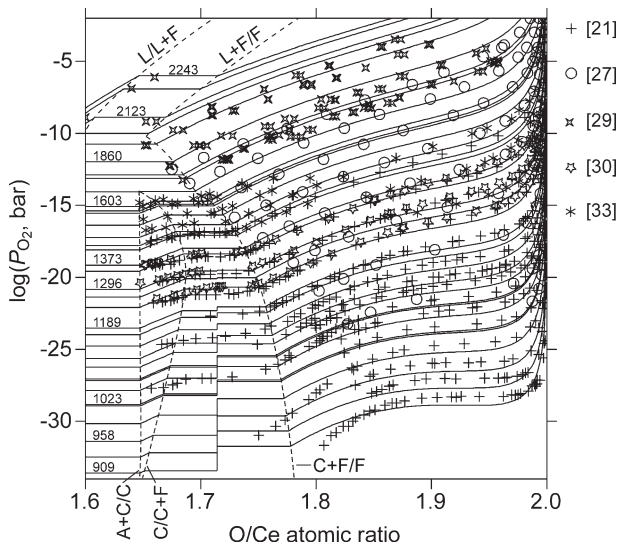

Fig. 3. Equilibrium oxygen pressures over the condensed phases in the Ce–O system versus composition (1.6<O/Ce<2.0). The curves are calculated for the following temperatures (from bottom to top, in K): 909, 933, 958, 988, 1020, 1023, 1050, 1070, 1073, 1098, 1117, 1145, 1173, 1189, 1244, 1273, 1296, 1353, 1373, 1426, 1442, 1473, 1522, 1573, 1583, 1603, 1673, 1750, 1773, 1860, 1998, 2081, 2123, 2203, 2243. Some selected curves are also labeled. The dashed lines show the phase boundaries.

As in the model for the A-phase, one of the charged parameters, ${}^{\circ}G^{C}_{Ce^{3+}:O^{2-}:O^{2-}}$ was again chosen as a reference, while the parameter ${}^{\circ}G^{C}_{Ce^{4+}:O^{2-}:Va}$ can be determined by the reciprocal relation. Similarly, the interaction parameters ${}^{v}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:O^{2-}}$ and ${}^{v}L^{C}_{Ce^{3+},Ce^{4+}:O^{2-}:Va}$ were used to model the homogeneity range of the C-phase, whereas ${}^{v}L^{C}_{Ce^{3+}:O^{2-}:O^{2-},Va}$ and ${}^{v}L^{C}_{Ce^{3+}:O^{2-}:O^{2-},Va}$ were not used and set equal to zero.

Stoichiometric ceria has a fluorite-like structure with two atomic positions in the unit cell, one for Ce^{4+} (4*a*) and one for O^{2-} (8*c*). When CeO_2 is reduced to CeO_{2-x} defects are formed

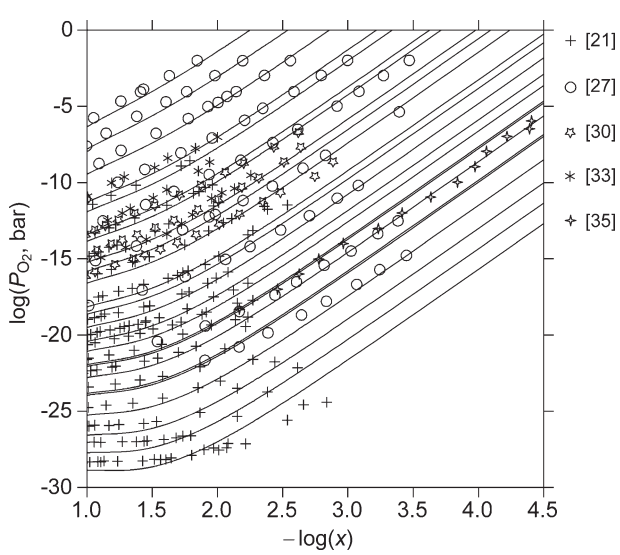

Fig. 4. Equilibrium oxygen pressures over CeO_{2-x} versus composition (x>1.9). The curves are calculated for the following temperatures (from bottom to top, in K): 909, 933, 958, 988, 1020, 1023, 1050, 1070, 1073, 1098, 1117, 1145, 1173, 1189, 1244, 1273, 1296, 1353, 1373, 1442, 1473, 1573, 1673, 1773.

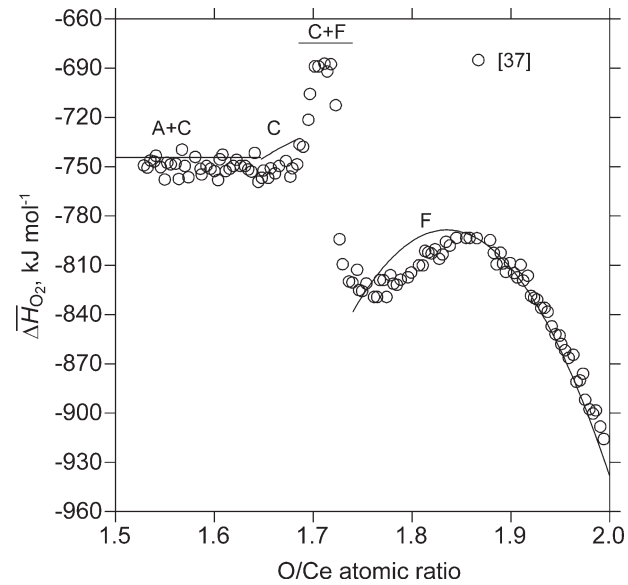

Fig. 5. Partial molar enthalpy of solution of O_2 in cerium oxides at 1353 K as function of composition. The corresponding phase fields are indicated.

in the form of oxygen vacancies and Ce^{3+}-cations [12]. The model for nonstoichiometric ceria can be expressed as $(Ce^{4+}, Ce^{3+})_2(O^{2-},Va)_4$ and can be visualized by the composition square in Fig. 1c. The molar Gibbs energy is then defined as

$$\begin{aligned} G^{F} = {} & y_{Ce^{4+}} y_{O^{2-}} {}^{\circ}G^{F}_{Ce^{4+}:O^{2-}} + y_{Ce^{4+}} y_{Va} {}^{\circ}G^{F}_{Ce^{4+}:Va} \\ & + y_{Ce^{3+}} y_{O^{2-}} {}^{\circ}G^{F}_{Ce^{3+}:O^{2-}} + y_{Ce^{3+}} y_{Va} {}^{\circ}G^{F}_{Ce^{3+}:Va} \\ & + 2RT(y_{Ce^{3+}} \ln y_{Ce^{3+}} + y_{Ce^{4+}} \ln y_{Ce^{4+}}) \\ & + 4RT(y_{O^{2-}} \ln y_{O^{2-}} + y_{Va} \ln y_{Va}) + {}^{E}G^{F} \end{aligned} \tag{10}$$

There is one neutral end-member ${}^{\circ}G^{F}_{Ce^{4+}:O^{2-}}$, which corresponds to the stoichiometric CeO_2. The other three end-members are charged, but a combination of 0.75 mol $(Ce^{3+})_2(O^{2-})_4$ and 0.25 mol $(Ce^{3+})_2(Va)_4$ gives the neutral compound

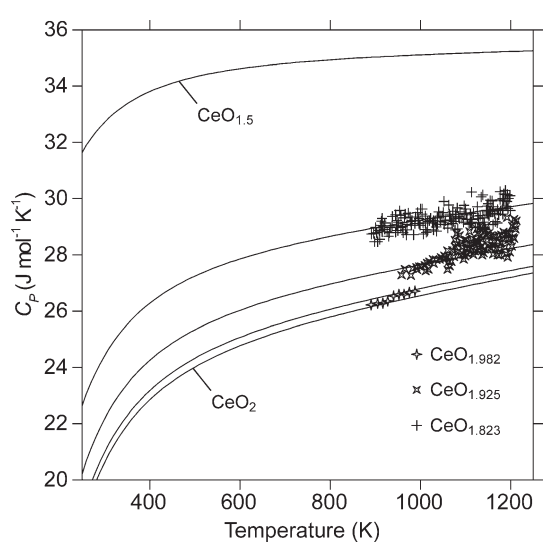

Fig. 6. Calculated heat capacity of the nonstoichiometric ceria as function of temperature superimposed with experimental points [57].

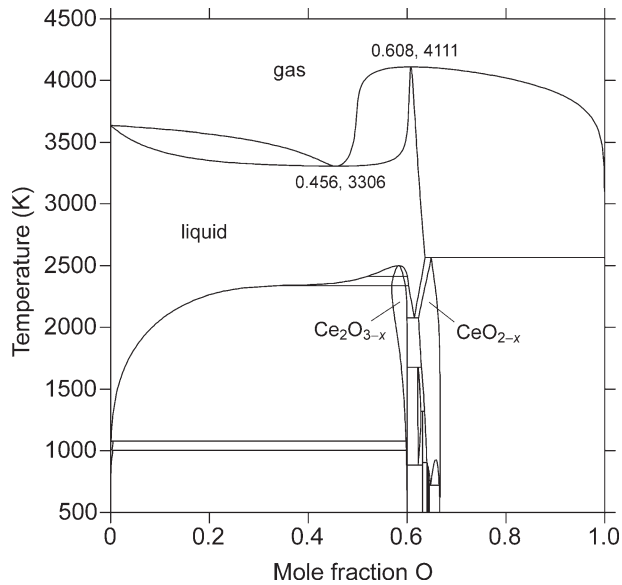

Fig. 7. Calculated phase diagram of Ce–O system for the total pressure of 1 bar.

$(Ce^{3+})_2(O^{2-})_3,(Va)_1$, which corresponds to Ce_2O_3 in the fluorite-related structure, i.e.

$$^{\circ}G_{F-Ce_2O_3} = \frac{3}{4}{}^{\circ}G^{F}_{Ce^{3+}:O^{2-}} + \frac{1}{4}{}^{\circ}G^{F}_{Ce^{3+}:Va} + 4RT\left(\frac{3}{4}\ln\frac{3}{4} + \frac{1}{4}\ln\frac{1}{4}\right), \quad (11)$$

where the last term is due to the ideal entropy of mixing on the anion sublattice. Using $^{\circ}G^{F}_{Ce^{4+}:Va}$ as a reference, i.e., $^{\circ}G^{F}_{Ce^{4+}:Va} = 2^{\circ}G_{CeO_2} - 2^{\circ}G_{O_2(g)}$ and the reciprocal relation

$$^{\circ}G^{F}_{Ce^{4+}:O^{2-}} + {}^{\circ}G^{F}_{Ce^{3+}:Va} = {}^{\circ}G^{F}_{Ce^{4+}:Va} + {}^{\circ}G^{F}_{Ce^{3+}:O^{2-}} \quad (12)$$

all the end-members in Eq. (10) can be defined. The term $^{\circ}G_{F-Ce_2O_3}$ was used together with the interaction parameters $^{\nu}L^{F}_{Ce^{3+},Ce^{4+}:O^{2-}}$ and $^{\nu}L^{F}_{Ce^{4+},Ce^{3+}:Va}$ to model the homogeneity range of CeO_{2-x}, whereas $^{\nu}L^{F}_{Ce^{4+}:O^{2-},Va}$ and $^{\nu}L^{F}_{Ce^{3+}:O^{2-},Va}$ were not used and set equal to zero.

4. Results and discussion

4.1. Thermodynamic functions for the stoichiometric Ce_2O_3 and CeO_2

The assessed Gibbs energy functions of both compounds are given in Table 4. The coefficients of Eq. (1) for A-Ce_2O_3 were obtained by fitting the heat capacity and enthalpy increment data [42,43,47], while using the standard entropy value of 148.8 J mol^{-1} K^{-1} [42]. The standard enthalpy of formation of the A-Ce_2O_3 was used as an adjustable parameter in the present work. The best fit of the phase diagram and thermodynamic data was obtained with $\Delta_f H^{\circ}$(298.15 K)= −1792.4 kJ mol^{-1}, which is well within the scatter of experimental measurements, −1788.0 to −1822.3 kJ mol^{-1} [38–42]. The enthalpies of A → H and H → X phase transitions in Ce_2O_3 and the entropy of melting as well as the Gibbs energy of the metastable C-Ce_2O_3 relative to the A-form were estimated [86] based on the pronounced correlation between thermodynamic properties of rare-earth sesquioxides and the cation radius, while the heat capacity of all Ce_2O_3 phases is described by the same function. There is a lack of corresponding measurements in the literature. All estimations are given in Table 4 together with other thermodynamic parameters describing the Ce–O system. The Gibbs energy equation for CeO_2 was obtained by fitting the corresponding heat capacity and enthalpy increment data [50,53–55,57], while accepting the standard entropy value of 62.3 J mol^{-1} K^{-1} [50] and the standard enthalpy of formation as −1090.4 kJ mol^{-1} [65]. The entropy of melting of CeO_2 (29.059 J mol^{-1} K^{-1}) was adopted from the SGTE substance database [87]. This value is very close to the entropy of melting of the isomorphous compound ZrO_2 [88], which also has a similar melting point. The enthalpy of melting of CeO_2 is then 87.177 kJ mol^{-1}. Again, the heat capacity of liquid CeO_2 is assumed to have the same temperature dependence, as the solid phase. The calculated heat capacity and enthalpy increment of the stoichiometric Ce_2O_3 and CeO_2 are shown in Fig. 2 in comparison with all available experimental data. It can be seen (Fig. 2a) that the heat capacity descriptions proposed in [45,51] are over-simplified and cannot be used in a wide temperature range. According to the present evaluation, the heat capacity of Ce_2O_3 is best represented by the data from [42,43], while that of CeO_2 by the data from [50,57]. The datapoints of [52] for CeO_2 show big deviations as already mentioned in the literature [89] and are actually off of the plot, whereas the other measurements for Ce_2O_3 [44,46] are slightly above the calculated curve. The calculated enthalpy increment of Ce_2O_3 (Fig. 2b) is in better agreement with the measurements of [42], while all three datasets for CeO_2 [53–55] show good consistency with the present assessment.

4.2. Solution phases and intermediate compounds

For the F- and the C-phase, interaction parameters of zeroth and first order (Table 4) were reliably determined based on the extensive experimental database (Table 2). Several datasets

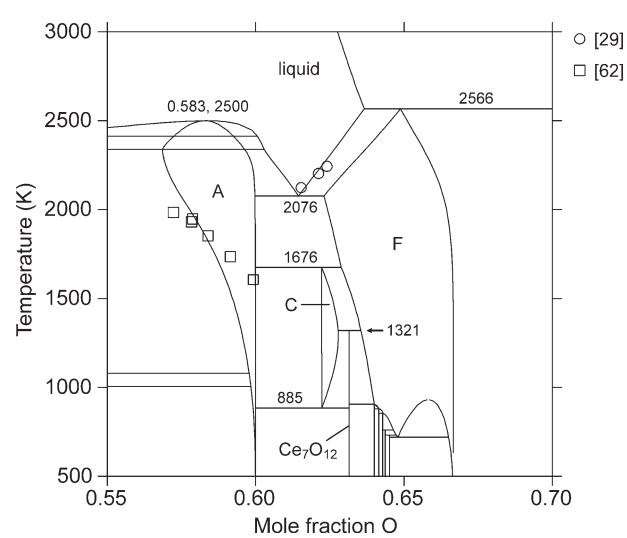

Fig. 8. The central part of Fig. 7 showing phase equilibria between cerium oxides.

Table 5
Invariant reactions in the Ce–O system

Reaction	Type	T (K)	ΔH_r (J mol^{-1})	Compositions of phases (at.% O)	Ref.
L=gas (P=1 bar)	Congruent	4111		60.8 (L, gas)	This work
L=gas (P=1 bar)	Congruent	3306		45.6 (L, gas)	This work
F=L+gas (P=1 bar)	Gas-peritectic	2566		63.7 (L), 64.9 (F)	This work
L=X	Congruent	2500		58.3 (L, X)	This work
L+X=H	Metatectic	2413		60.1 (L), 59.3 (X, H)	This work
L+X=H	Metatectic	2413		52.0 (L), 57.2 (X, H)	This work
L+H=A	Metatectic	2338		60.3 (L), 59.6 (H, A)	This work
L+H=A	Metatectic	2338		33.7 (L), 56.9 (H, A)	This work
L=A+F	Eutectic	2076		61.4 (L), 60.0 (A), 62.3 (F)	This work
A+F=C	Peritectoid	1676		60.0 (A), 62.9 (F), 62.2 (C)	This work
C+F=Ce_7O_{12}	Peritectoid	1321		62.8 (C), 63.5 (F)	This work
		1296			[21]
		1084			[56]
		1072			[15]
C=A+Ce_7O_{12}	Eutectoid	885		62.2 (C), 60.0 (A)	This work
		873			[21]
Ce_7O_{12}+F=Ce_9O_{16}	Peritectoid	905		64.003 (F)	This work
		952			[60]
		913			[56]
		905		64.1 (F)	[61]
		911			[15]
Ce_9O_{16}+F=$Ce_{19}O_{34}$	Peritectoid	880		64.154 (F)	This work
		880			[56]
		898			[15]
$Ce_{19}O_{34}$+F=$Ce_{40}O_{72}$	Peritectoid	855		64.310 (F)	This work
		850			[56]
		860		64.5 (F)	[61]
		866			[15]
$Ce_{40}O_{72}$+F=$Ce_{62}O_{112}$	Peritectoid	760	−94	64.647 (F)	This work
		766	−200		[56]
		756		64.7 (F)	[61]
		755			[15]
$Ce_{62}O_{112}$+F=$Ce_{11}O_{20}$	Peritectoid	732	−183	64.736 (F)	This work
		736	−229		[56]
		725		64.8 (F)	[61]
F′=F″+$Ce_{11}O_{20}$	Eutectoid	721	−473	64.802 (F′), 66.504 (F″)	This work
		722	−565	64.86 (F′), 66.60 (F″)	[56]
		714			[61]
F=F′+F″	Congruent	930		65.860 (F, F′, F″)	This work
L+A=bcc-Ce	Peritectic	1080		0.11 (L), 59.8 (A), 0.57 (bcc)	This work
bcc-Ce+A=fcc-Ce	Peritectoid	1005		0.31 (bcc), 59.8 (A), 0.49 (fcc)	This work

The enthalpies of reactions (ΔH_r) refer to one mole of atoms.

were not included in the optimization either because they largely disagree with the majority of reports [23,25,26,28] or because not the original data, but smoothed curves have been presented [24,34,36]. In addition, heat capacity data measured for the compositions $CeO_{1.982}$, $CeO_{1.925}$, and $CeO_{1.823}$ allowed to evaluate the end-member $^{\circ}G_{F-Ce_2O_3}$ in the form of Eq. (1), while the end-member $^{\circ}G^{C}_{Ce^{4+}:O^{2-}:O^{2-}}$ was expressed relative to $^{\circ}G_{CeO_2}$ (Table 4) to avoid unexpected stabilisation of the C-phase in the F-region. The Gibbs energy of the intermediate cerium oxides (i.e., $Ce_{11}O_{20}$, $Ce_{62}O_{112}$, $Ce_{40}O_{72}$, $Ce_{19}O_{34}$, Ce_9O_{16}, Ce_7O_{12}) was evaluated relative to $^{\circ}G_{CeO_2}$ and $^{\circ}G_{F-Ce_2O_3}$ according to the stoichiometry in the form $a+bT$. This implies that the heat capacity of those compounds is set equal to the heat capacity of the F-phase of the same composition, since the corresponding experimental data are lacking.

For the A-phase, only a regular interaction parameter was evaluated. This is sufficient to reproduce the available experimental data on the oxygen substoichiometry [62]. For the interstitial solid solution phases, fcc- and bcc-Ce, only the end-members, $^{\circ}G^{fcc}_{Ce:O}$ and $^{\circ}G^{bcc}_{Ce:O}$ were evaluated relative to pure elements (Table 4) based on the measurements of oxygen solubility in solid cerium [63].

For the liquid phase, the adjustable parameters were temperature-independent terms $^{\nu}L^{liq}_{Ce^{3+},Ce^{4+}:O^{2-}}, \nu = 0, 1$ and $^{0}L^{liq}_{Ce^{3+}:O^{2-},Va^{q-}}$ as well as the enthalpy of melting of Ce_2O_3 (Table 4), since the congruent melting point does not exactly correspond to O/Ce=1.5 (see below). They were evaluated simultaneously using data on the oxygen solubility in liquid cerium [63] and the liquidus points in the Ce_2O_3–CeO_2 region [29].

The calculated equilibrium oxygen pressures for the composition range O/Ce=1.6–2.0 are shown in Fig. 3 in comparison with selected experimental data [21,27,29,30,33,35]. The temperatures, for which oxygen pressures are calculated are taken from those publications. The measured oxygen pressures

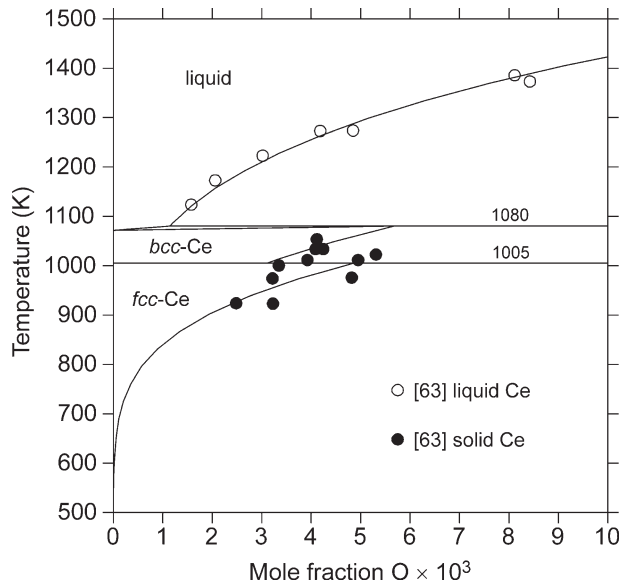

Fig. 9. The cerium-rich part of Fig. 7 showing phase equilibria between liquid, interstitial solution phases and solid Ce_2O_3.

are reproduced within the limits of experimental errors. The low-temperature measurements of Bevan and Kordis [21] deviate from the curves with decreasing oxygen content (Fig. 3) because equilibrium was not established as pointed out by the authors themselves. Good agreement was also found to exist between calculations and data from [19,20,22,31,32], which are not included in Fig. 3 to ensure clarity. The plateaus correspond to the miscibility gap in the nonstoichiometric ceria around O/Ce = 1.9 and to the biphasic regions F + Ce_7O_{12}, F + C, F + A, F + liquid, Ce_7O_{12} + C, and C + A, while the curves partly overlap at the composition of Ce_7O_{12}.

Fig. 4 shows equilibrium oxygen pressures for the composition range O/Ce > 1.9. Note that the composition axis here is logarithmic. It is worth mentioning that traditional defect chemistry predicts a straight line with a slope of −1/6 for the

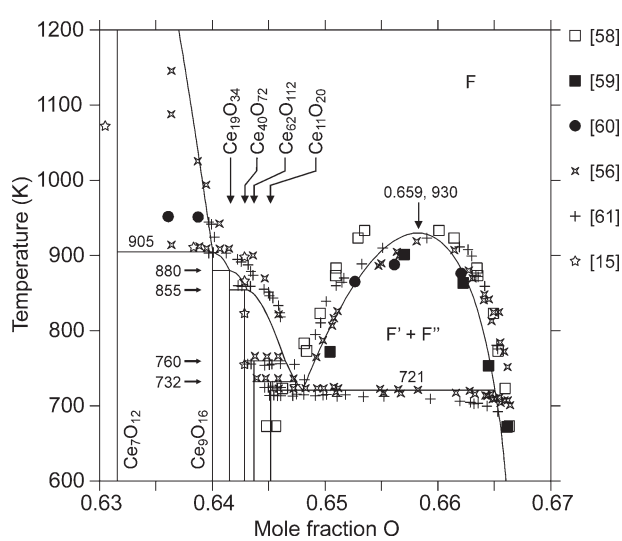

Fig. 10. A part of Ce–O phase diagram between Ce_7O_{12} and CeO_2 at low temperatures.

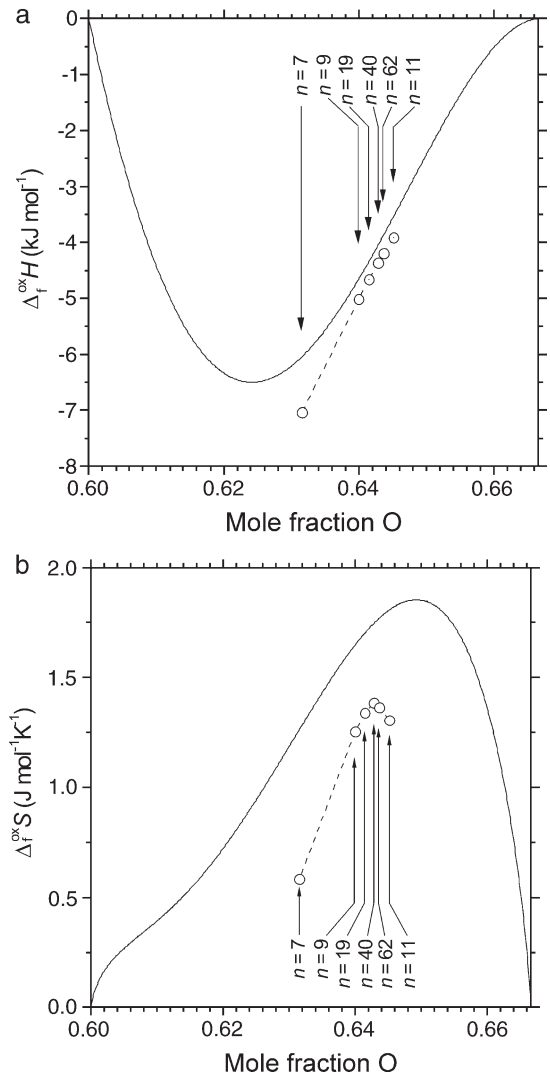

Fig. 11. Enthalpy (a) and entropy (b) of formation of Ce_nO_{2n-2m} compounds from CeO_2 and fluorite-like Ce_2O_3 (circles) in comparison with the enthalpy and entropy of mixing in the nonstoichiometric ceria (solid curves) at 298.15 K. All values refer to one mole of atoms. The dashed lines are only guides for eye.

plot of $\log(P_{O2})$ versus $\log(x)$. As it can be seen in Fig. 4, this approach is accurate enough as long as the deviations from stoichiometry in CeO_{2-x} are very small ($x > 1.99$) but fails at higher defect concentrations. At low temperatures the slope even reaches the value zero when passing through the miscibility gap, then increases again. The present modelling uses the same basic assumptions as the traditional defect chemistry, but the *non-ideality* is introduced and expressed quantitatively in terms of the interaction parameters ${}^0L^F_{Ce^{3+},Ce^{4+}:O^{2-}}$ and ${}^1L^F_{Ce^{3+},Ce^{4+}:O^{2-}}$. In this way, the change of slope is described very well without a need to introduce artificial clusters to model the nonstoichiometry of the CeO_{2-x} phase. In addition, since the fraction of Ce^{3+} ions in the cation sublattice of CeO_{2-x} is always equal to $2x$, Figs. 3 and 4 can be used to obtain the concentration of small polarons (which are responsible for the electronic conductivity of ceria [12]) at different temperatures and oxygen pressures. This is important to estimate the applicability of ceria-based materials as solid electrolytes.

Fig. 5 presents a plot of calculated partial molar enthalpy of solution of O_2 in cerium oxides at 1353 K in comparison with the measurements [37]. The agreement is quite satisfactory, except for the compositions adjacent to biphasic region F+C. Since the measurements were done by stepwise oxidation of the sample it is believed that the heat effect registered in the regions O/Ce=1.69–1.70 and O/Ce=1.72–1.75 reflect a transient state rather than the true equilibrium. Indeed, $\overline{\Delta H}^{O_2}$ must show a discontinuity when a phase boundary is crossed and the corresponding gaps between calculated curves are clearly seen in Fig. 5. Panhans and Blumenthal [35] suggested that at 1073 K, $\overline{\Delta H}O_2$ goes through a minimum around $CeO_{1.998}$ and reaches the value of −500 kJ·mol^{-1} at the near-stoichiometric composition $CeO_{1.999997}$. This trend is not confirmed in the present work. It should be noted that the enthalpy was not measured directly in [35] and the accurate measurements of extremely small deviations from the stoichiometry in CeO_2 over a range of temperatures are still lacking in the literature. In Fig. 6, the calculated heat capacity of CeO_{2-x} at several compositions (normalized to one mole of atoms) is compared with experimental data [57]. A very good consistency is evident. It can be also seen that the heat capacity of the fluorite-like Ce_2O_3 (if it would exist) would be very high.

Fig. 7 shows the phase diagram of the Ce–O system calculated in the temperature range 500–4500 K for the total pressure of 1 bar. The liquid phase extends over a wide range of compositions and shows a tendency for demixing around 35 at.% O, where the corresponding liquidus curve becomes very flat.

The central part of the diagram (Fig. 8) is occupied by the A-, C-, F-, and Ce_nO_{2n-2m} phases. The former melts congruently at slightly substoichiometric composition, whereas ceria decomposes on heating into the liquid phase and oxygen gas. It is interesting that the temperature of this gas-peritectic reaction (2566 K) is rather close the "melting point of CeO_2" reported in [67,74] (Table 3). Two horizontal lines just below the congruent melting point of X-phase in Fig. 8, correspond to the polymorphic transformations in Ce_2O_3, X=H and H=A, respectively. It can be seen that a very good fit to the liquidus data [29] was obtained, while the calculated slope of the liquid+A/A phase boundary is slightly smaller than reported in [62]. Attempts to fit the experimental slope resulted in an artifact that the A-phase became stable at the composition Ce_2O_2. It is concluded that the specimens in [62] were probably not fully equilibrated at lower temperatures. The boundaries of the F- and C-phases correspond to the points of intersection between horizontal lines and curves in Fig. 3 and show good compliance with experimental data. The temperatures of the invariant reactions C+F=Ce_7O_{12} and C=A+Ce_7O_{12} were calculated as 1321 and 885 K, respectively (Table 5). They are in close agreement with the results of [21]. Two other works [15,56] indicated significantly lower temperature for the former reaction. However, it is just one point in [56] that may be associated with the decomposition of Ce_7O_{12}, while no X-ray diffraction analysis of the samples was made. Kümmerle et al. [15] reported that Ce_7O_{12} transformed directly to F, what is in contradiction with the phase diagram (Fig. 8). It is therefore believed that the temperature between 1072 and 1084 K corresponds to the rapid oxidation of the sample, so that single F-phase is formed. The stability of Ce_7O_{12} at higher temperatures is also supported by the measurements of equilibrium oxygen pressure versus composition at 1244 and 1296 K [30], where the clear plateaus, corresponding to the equilibria with Ce_7O_{12} were observed (cf. Fig. 3). In Fig. 9, the calculated metal-rich part of the Ce–O phase diagram is shown. One can see that experimental data [63] are reproduced with high accuracy.

Fig. 10 shows phase equilibria in the Ce–O system in the region between Ce_7O_{12} and CeO_2 at low temperatures. It is evident that the present calculation complies well with most experimental information. Comparing Figs. 3 and 10 it can be concluded that the equilibrium oxygen pressure measurements and phase diagram data, which concern the miscibility gap in the F-phase are in good accord. The invariant peritectoid/eutectoid reactions, in which Ce_nO_{2n-2m} and F-phases are involved are listed in Table 5. The best agreement exists with the data of [56]. The enthalpies of those transformation, where they were measured are reproduced within the limits of experimental error. The transition temperature of 823 K [15], however, does not correspond to any invariant reaction in the present assessment. It may be related to the decomposition of one of the Ce_nO_{2n-2m} compounds with n=13, 29 or 39 [11], which were not taken into account in the present work. The possible existence of such phases may also be inferred from the largest deviations between calculated and measured enthalpy of the peritectoid reaction $Ce_{40}O_{72}$+F=$Ce_{62}O_{112}$ (Table 5).

It is now interesting to compare the enthalpy and entropy of formation of Ce_nO_{2n-2m} phases relative to F solid solution (Fig. 11). The enthalpy of mixing in the F-phase shows a minimum around 62.4 at.% O, whereas the entropy of mixing passes through the maximum around 64.9 at.% O. The enthalpy and entropy of formation of Ce_nO_{2n-2m} exhibit similar behaviour: enthalpy decreases from $Ce_{11}O_{20}$ to Ce_7O_{12}, while the entropy reaches its maximum value for $Ce_{40}O_{72}$. The observed dependencies are quite logical in a sense that all the discrete compounds Ce_nO_{2n-2m} are superstructures of the fluorite-type lattice. In addition, Fig. 11 shows that extrapolating the composition dependence of $\Delta_f^{ox}S°$ would result in a zero or even negative value at 62.5 at.% O (Fig. 11b). This can explain, why the stoichiometric compound Ce_3O_5 (n=6, m=1) is not stable at low temperatures, but exists as a solution phase (C) due to the positive entropy of mixing.

5. Concluding remarks

The experimental database for the Ce–O system is large and difficult to use for practical purposes. In the present work, this information was condensed in the form of coefficients of the Gibbs energy equations, which are parts of thermodynamic models for individual phases. The models developed in this work are, on the one hand, simple to be compatible with the standard Gibbs energy minimization procedure. On the other hand, they provide enough flexibility to account for the crystal structure, defect chemistry, and thermodynamic properties of ionic solids. Using the parameters of thermodynamic description

of the Ce–O system evaluated in the present work (Table 4) any thermochemical property and a phase diagram can be calculated. This is a significant improvement compared to the previous modelling works, which have dealt mainly with the CeO_{2-x} phase. With a few exceptions [23,25,26,28,52] a good consistency of the experimental database (Table 2) was demonstrated. While present work accumulates all previous knowledge about the Ce–O system, several new results were also obtained:

1) Probable value for the melting point of the stoichiometric CeO_2 (3000±20 K). The melting is congruent only under elevated pressure.
2) The Ce–O phase diagram in the whole range of compositions (Fig. 7) or at elevated temperature (Fig. 8) not reported previously.
3) A complete list of invariant reactions in the Ce–O system (Table 5). While not all these temperatures and compositions have been measured, the calculated values are based on the real thermodynamic and phase boundary data and thus, contribute to the database.
4) The enthalpy and entropy of formation of Ce_nO_{2n-2m} phases (Fig. 11) assessed for the first time.

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